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Separating Differential Equations : Sedra Khan

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a) $y' = f(x)g(y) \Rightarrow \frac{dy}{dx} = f(x)g(y)$

rewrite
 $\therefore \frac{dy}{g(y)} = f(x)dx$, next we integrate

$$\int \frac{1}{g(y)} dy = \int f(x) dx \Rightarrow S = K + C_1$$

b) $y' + p(x)y = Q(x)$ LINEAR $\therefore \mu(x) = e^{\int p(x)dx + K}$
 $\therefore \mu(x) = e^{\int p(x)dx}$

LHS $\rightarrow \mu(x)y' + \mu(x)p(x)y = \mu(x)Q(x)$
 $\rightarrow [(\mu(x) \cdot y)]' = \mu(x)Q(x)$

integrate $\rightarrow \mu(x) \cdot y = \int \mu(x) \cdot Q(x) + C$
 $y = \int \mu(x) \cdot Q(x) + \frac{C}{\mu(x)}$

$$y = \frac{\int e^{\int p(x)dx} \cdot Q(x) + \frac{C}{e^{\int p(x)dx}}}$$

c) $y'' + b^2y = 0$ 2nd ODE

guess: $y = e^{rx}$, $y' = r e^{rx}$, $y'' = r^2 e^{rx}$
Substituting these guesses into the original eqn:

$$r^2 e^{rx} + b^2 e^{rx} = 0 \rightarrow \frac{e^{rx}(r^2 + b^2)}{e^{rx}} = 0$$

$\therefore r^2 + b^2 = 0$
 $r^2 = -b^2 \rightarrow r = \pm bi$

$y = e^0 (C_1 \cos(bx) + C_2 \sin(bx))$

$\therefore y = C_1 \cos(bx) + C_2 \sin(bx)$

d) $y'' - b^2y = 0$ 2nd ODE

Same guesses as (c)

$$r^2 e^{rx} - b^2 e^{rx} = 0 \rightarrow \frac{e^{rx}(r^2 - b^2)}{e^{rx}} = 0$$

$\therefore r^2 - b^2 = 0$
 $r^2 = b^2 \rightarrow r = \pm b$

$y = C_1 e^{bx} + C_2 e^{-bx}$

e) $y'' = b_0 + b_1x + b_2x^2$

$$\int \frac{dy^2}{dx^2} = \int b_0 + b_1x + b_2x^2$$

$$\frac{dy}{dx} = b_0x + b_1 \frac{x^2}{2} + b_2 \frac{x^3}{3} + C$$

$\int dy = \int b_0x + b_1 \frac{x^2}{2} + b_2 \frac{x^3}{3} + C dx$

$$y = b_0 \frac{x^2}{2} + b_1 \frac{x^3}{6} + b_2 \frac{x^4}{12} + C_1x + C_2$$

f) $\frac{1}{x} \frac{d}{dx} (x \frac{dy}{dx}) = b_0 + b_1x + b_2x^2$

$$\frac{d}{dx} (x \frac{dy}{dx}) = b_0x + b_1x^2 + b_2x^3$$

integrate, in terms of x
 $x \frac{dy}{dx} = \frac{b_0x^2}{2} + \frac{b_1x^3}{3} + \frac{b_2x^4}{4} + C_1$

$$\frac{dy}{dx} = \frac{b_0x}{2} + \frac{b_1x^2}{3} + \frac{b_2x^3}{4} + \frac{C_1}{x}$$

integrate on both ends.
 $y = \frac{b_0x^2}{4} + \frac{b_1x^3}{9} + \frac{b_2x^4}{16} + C_1 \ln|x| + C_2$

g) $\frac{1}{x^2} \frac{d}{dx} (x^2 \frac{dy}{dx}) + b^2y = 0$ Sub $y(x) = \frac{u(x)}{x}$

$\therefore \frac{u(x)}{x} = x \cdot y$
 $\frac{dy}{dx} = -x^{-2}u + u'x^{-1}$

$$\frac{1}{x^2} \frac{d}{dx} (x^2 (-\frac{u}{x^2} + \frac{u'}{x})) + b^2 \frac{u}{x} = 0$$

$$\frac{1}{x^2} \frac{d}{dx} (-u + u'x) + b^2 \frac{u}{x} = 0$$

$$\frac{1}{x^2} (-u' + u' + u''x) + b^2 \frac{u}{x} = 0$$

$$\frac{u''x + b^2u \cdot x}{x} = 0$$

$u'' + b^2u = 0$
 $u = \sqrt{-b^2} = -bi$
 $u = C_1 \cos bx + C_2 \sin bx$

$$y = \frac{\cos bx \cdot C_1 + \sin bx \cdot C_2}{x}$$

h) $\frac{1}{x^2} \frac{d}{dx} (x^2 \frac{dy}{dx}) - b^2y = 0$ again, $y = \frac{u(x)}{x}$

$$\frac{1}{x^2} \frac{d}{dx} (x^2 (-\frac{u}{x^2} + \frac{u'}{x})) - b^2 \frac{u}{x} = 0$$

$$\frac{1}{x^2} (-u' + u' + u''x) - b^2 \frac{u}{x} = 0 \rightarrow \frac{u''x - b^2u \cdot x}{x} = 0$$

$u'' - b^2u = 0$
 $u = \pm b$
 $y = \frac{C_1 e^{bx} + C_2 e^{-bx}}{x}$

6) $y'' + b_1 y' + b_2 y = 0$

$r^2 e^{rx} + b_1 r e^{rx} + b_2 e^{rx} = 0$

$r^2 + b_1 r + b_2 = 0 \rightarrow -b_1 \pm \sqrt{b_1^2 - 4b_2}$

let's call
the discriminant,

"m"

assuming that
the discriminant
is greater than 0

$y = C_1 e^{\frac{(-b_1 + m)x}{2}} + C_2 e^{\frac{(-b_1 - m)x}{2}}$

cases: $\Delta < 0$ &
 $\Delta = 0$
missing
(-3)

7) $y'' + 2xy' = 0$ POWER SERIES

$y'' = -2xy'$

let's call this u
 $y'' = u'$

$u' = -2x \cdot u$

integrate $\rightarrow u = \int -2x \cdot u$

$u = e^{-x^2} + C_1$

$y' = e^{-x^2} + C_1$

integrate,
once more.

$y = e^{C_1} \frac{\sqrt{\pi}}{2} \text{erf}(x) + C_2$

which can also be known,

$y = C_1 \frac{\sqrt{\pi}}{2} \text{erf}(x) + C_2$

8) $\frac{d}{dx} [f(y) \frac{dy}{dx}] = 0$

$f(y) \frac{dy}{dx} = \int 0$

$f(y) \frac{dy}{dx} = 0 + C_1$

$f(y) \frac{dy}{dx} = C_1$

$\int f(y) \frac{dy}{dx} = C_1 x + C_2$

$S = C_1 x + C_2$

9) $\frac{1}{x^2} \frac{d}{dx} (x^2 \frac{dy}{dx}) = b_0 + b_1 x + b_2 x^2$

$\frac{d}{dx} (x^2 \frac{dy}{dx}) = b_0 x^2 + b_1 x^3 + b_2 x^4$

$x^2 \frac{dy}{dx} = \frac{b_0 x^3}{3} + \frac{b_1 x^4}{4} + \frac{b_2 x^5}{5} + C_1$

$\frac{dy}{dx} = \frac{b_0 x}{3} + \frac{b_1 x^2}{4} + \frac{b_2 x^3}{5} + \frac{C_1}{x^2}$

$y = \frac{b_0 x^2}{6} + \frac{b_1 x^3}{12} + \frac{b_2 x^4}{20} - C_1 x^{-1} + C_2$